CS 7/5382 - HW 3: Step-by-Step Solutions

Name: Daniel Oni

Problem 1 (30 points) - Rays, Lines, and Intersections

Given: A = [1, 1, 1, 1] and B = [2, -1, 3, 1] in 4D homogeneous coordinates.

1a) Define a ray r that passes through A in the direction of B

A ray is defined as $P(\alpha) = Q + \alpha\mathbf{d}$, where Q is a starting point and **d** is a direction vector.

Step 1: Compute direction vector **d** = B - A (point minus point = vector):

$$d = B - A = [2-1, -1-1, 3-1, 1-1] = [1, -2, 2, 0]$$

Note: The w-component is 0, confirming this is a vector (not a point) in homogeneous coordinates.

Step 2: Write the ray (restricting $\alpha \geq 0$):

$$\mathbf{r(\alpha) = A + \alpha d = [1, 1, 1, 1] + \alpha[1, -2, 2, 0],\ \alpha \geq 0}$$

1b) Express the line through A and B in parametric form

The parametric line uses the same formula but allows α to range over all real numbers:

$$P(\alpha) = A + \alpha(B - A) = [1, 1, 1, 1] + \alpha[1, -2, 2, 0]$$

$$\mathbf{x(\alpha) = 1 + \alpha}$$

$$\mathbf{y(\alpha) = 1 - 2\alpha}$$

$$\mathbf{z(\alpha) = 1 + 2\alpha}$$

$$\mathbf{w(\alpha) = 1}$$

$$\mathbf{for\ all\ real\ \alpha}$$

1c) Sketch point A and ray r; show where line intersects the XZ-plane

The XZ-plane is defined by y = 0. From the parametric form:

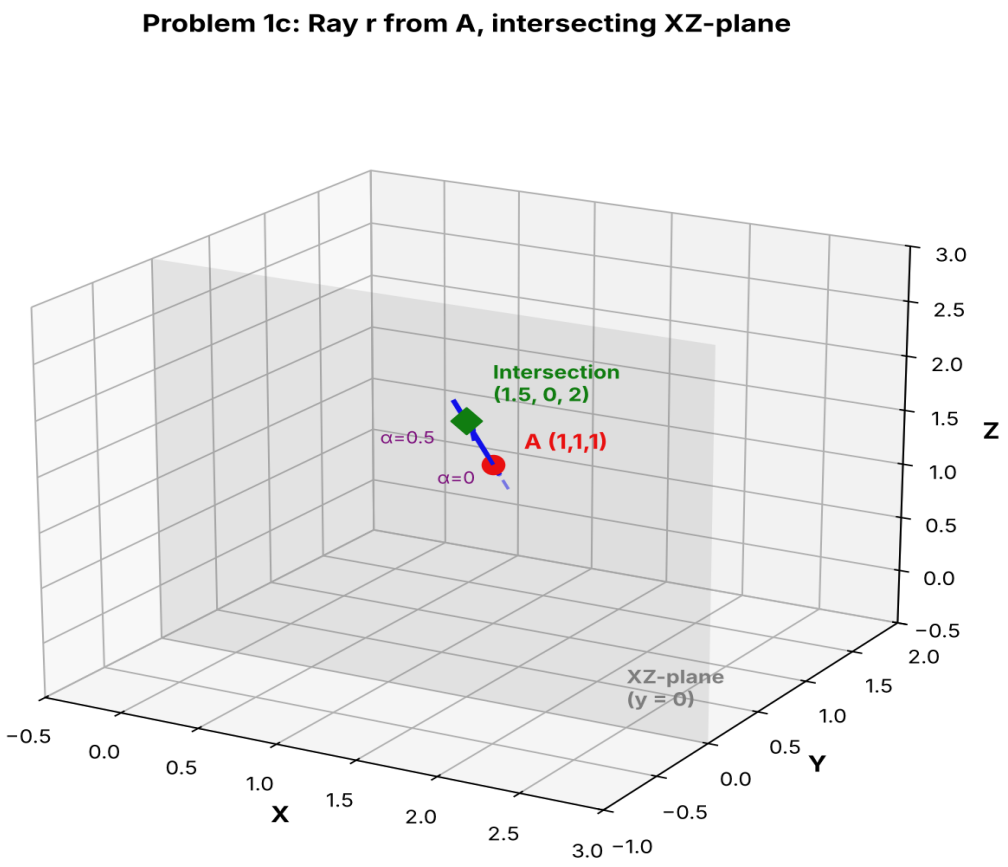
Step 1: Set $y(\alpha) = 0$ and solve for α:

$$1 - 2\alpha = 0 \rightarrow \alpha = 1/2 = 0.5$$

Step 2: Since $\alpha = 0.5 > 0$, the intersection lies on the ray.

Step 3: Compute the intersection point:

$$x(0.5) = 1.5,\ y(0.5) = 0,\ z(0.5) = 2$$



Point A sits above the XZ-plane at (1,1,1). The ray travels in direction [1,-2,2], passing through the XZ-plane at (1.5, 0, 2) where $\alpha = 0.5$.

1d) Express the XZ-plane intersection in homogeneous coordinates

`Intersection = [1.5, 0, 2, 1] or equivalently [3/2, 0, 2, 1]`

(w = 1 because this is a point.)

Problem 2 (20 points) - Transformation Ordering

Given: Horizontal translation of 8 (t_x=8) and rotation about Z-axis of 30°.

For 30°: cos(30°) = √3/2 ≈ 0.866, sin(30°) = 1/2 = 0.5

Building Block Matrices:

Translation matrix T(8, 0, 0):

```
| 1 0 0 8 |
T = | 0 1 0 0 |
| 0 0 1 0 |
| 0 0 0 1 |
```

Rotation matrix R_z(30°):

```
| cos30 -sin30 0 0 | | 0.866 -0.5 0 0 |
Rz = | sin30 cos30 0 0 | = | 0.5 0.866 0 0 |
| 0 0 1 0 | | 0 0 1 0 |
| 0 0 0 1 | | 0 0 0 1 |
```

2a) Construct M_A (translate first, then rotate)

In v' = M_Av, the matrix closest to v acts first. Translate first, then rotate:

```
M_A = Rz(30) * T(8,0,0)
```

Multiply R_z(30°) × T(8,0,0):

```
| 0.866 -0.5 0 6.928 |
M_A = | 0.5 0.866 0 4.0 |
| 0 0 1 0 |
| 0 0 0 1 |
```

(The translation column: 8×cos30° = 6.928, 8×sin30° = 4.0 — the translation itself got rotated.)

2b) Construct M_B (rotate first, then translate)

Rotate first, then translate:

```
M_B = T(8,0,0) * Rz(30)
```

```
| 0.866 -0.5 0 8 |
M_B = | 0.5 0.866 0 0 |
| 0 0 1 0 |
```

```
| 0 0 0 1 |
```

Key difference: In M_A, the translation values are rotated (6.928, 4.0). In M_B, the translation stays along x-axis (8, 0) because it is applied after the rotation.

2c) Apply each transformation to the point (-1, 2)

Express as homogeneous coordinate: p = $[-1, 2, 0, 1]^T$

Apply M_A:

```
x' = 0.866(-1) + (-0.5)(2) + 0(0) + 6.928(1)
= -0.866 - 1.0 + 6.928 = 5.062
y' = 0.5(-1) + 0.866(2) + 0(0) + 4.0(1)
= -0.5 + 1.732 + 4.0 = 5.232
```

M_A * [-1, 2, 0, 1]^T = [5.062, 5.232, 0, 1]^T

Apply M_B:

```
x' = 0.866(-1) + (-0.5)(2) + 0(0) + 8(1)
= -0.866 - 1.0 + 8.0 = 6.134
y' = 0.5(-1) + 0.866(2) + 0(0) + 0(1)
= -0.5 + 1.732 = 1.232
```

M_B * [-1, 2, 0, 1]^T = [6.134, 1.232, 0, 1]^T

Comparison: M_A (translate then rotate) moved the point to (5.062, 5.232) — orbits far from origin. M_B (rotate then translate) moved it to (6.134, 1.232) — rotates in place, then shifts right. Transformation order matters!

Problem 3 (50 points) - Concatenated Transformations

Given: q' = R_z(90°) S(2, 4, 1) q

3a) Rewrite as concatenation of two 4x4 matrices

cos(90°) = 0, sin(90°) = 1

S(2, 4, 1):

```
| 2 0 0 0 |
| 0 4 0 0 |
| 0 0 1 0 |
| 0 0 0 1 |
```

R_z(90°):

```
| 0 -1 0 0 |
| 1 0 0 0 |
| 0 0 1 0 |
| 0 0 0 1 |
```

q' = Rz(90) * S(2,4,1) * q

3b) Multiply to get single equivalent matrix M

M = R_z(90°) × S(2, 4, 1). Multiply row-by-column:

```
Row 1 of Rz * Cols of S:
[0,-1,0,0]*col1=[0,-1,0,0].[2,0,0,0]=0
[0,-1,0,0]*col2=[0,-1,0,0].[0,4,0,0]=-4
[0,-1,0,0]*col3=[0,-1,0,0].[0,0,1,0]=0
[0,-1,0,0]*col4=[0,-1,0,0].[0,0,0,1]=0
Row 2 of Rz * Cols of S:
[1,0,0,0]*col1=2, col2=0, col3=0, col4=0
Rows 3 and 4: identity portion unchanged
```

M =

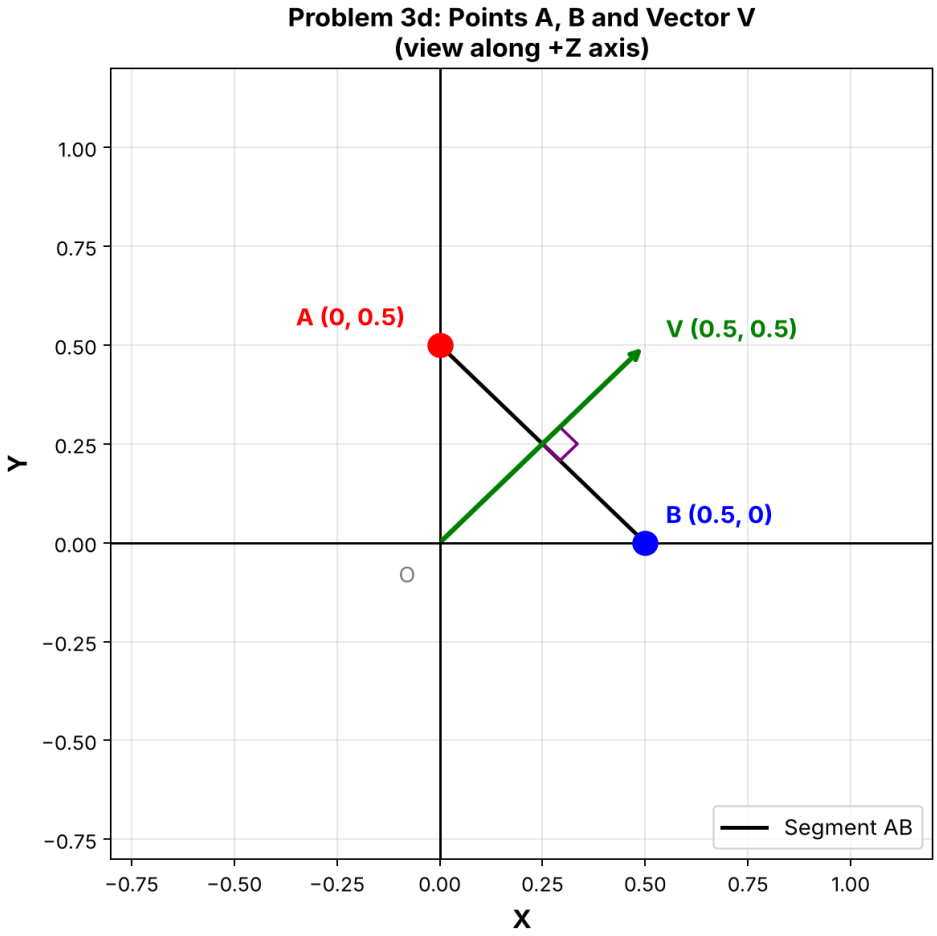
```
| 0 -4 0 0 |
| 2 0 0 0 |
| 0 0 1 0 |
| 0 0 0 1 |
```

3c) Express points and vector as 4x1 homogeneous coordinates

Points have w = 1, vectors have w = 0:

```
Point A: (0.0, 0.5) --> [0.0, 0.5, 0.0, 1]^T
Point B: (0.5, 0.0) --> [0.5, 0.0, 0.0, 1]^T
Vector V: (0.5, 0.5) --> [0.5, 0.5, 0.0, 0]^T
```

3d) Sketch and label all elements (view along +Z axis)



A is at (0, 0.5) on the Y-axis. B is at (0.5, 0) on the X-axis. Segment AB connects them. Vector V = (0.5, 0.5) points at 45° from the origin. The small square at the midpoint of AB indicates that V is perpendicular to AB.

3e) Is vector V normal to line segment AB?

Step 1: Direction vector of AB:

```
AB = B - A = [0.5-0.0, 0.0-0.5] = [0.5, -0.5]
```

Step 2: Dot product V · AB:

```
V . AB = (0.5)(0.5) + (0.5)(-0.5) = 0.25 - 0.25 = 0
```

`Since V . AB = 0, vector V IS normal (perpendicular) to segment AB.`

Two vectors are perpendicular if and only if their dot product equals zero.

3f) Apply transformation M to points A, B and vector V

Using M from part (b):

```
| 0 -4 0 0 |
M = | 2 0 0 0 |
| 0 0 1 0 |
| 0 0 0 1 |
```

A' = M × A:

```
x' = 0(0) + (-4)(0.5) + 0(0) + 0(1) = -2.0
y' = 2(0) + 0(0.5) + 0(0) + 0(1) = 0.0
```

A' = [-2.0, 0.0, 0.0, 1]^T

B' = M × B:

```
x' = 0(0.5) + (-4)(0) + 0(0) + 0(1) = 0.0
y' = 2(0.5) + 0(0) + 0(0) + 0(1) = 1.0
```

B' = [0.0, 1.0, 0.0, 1]^T

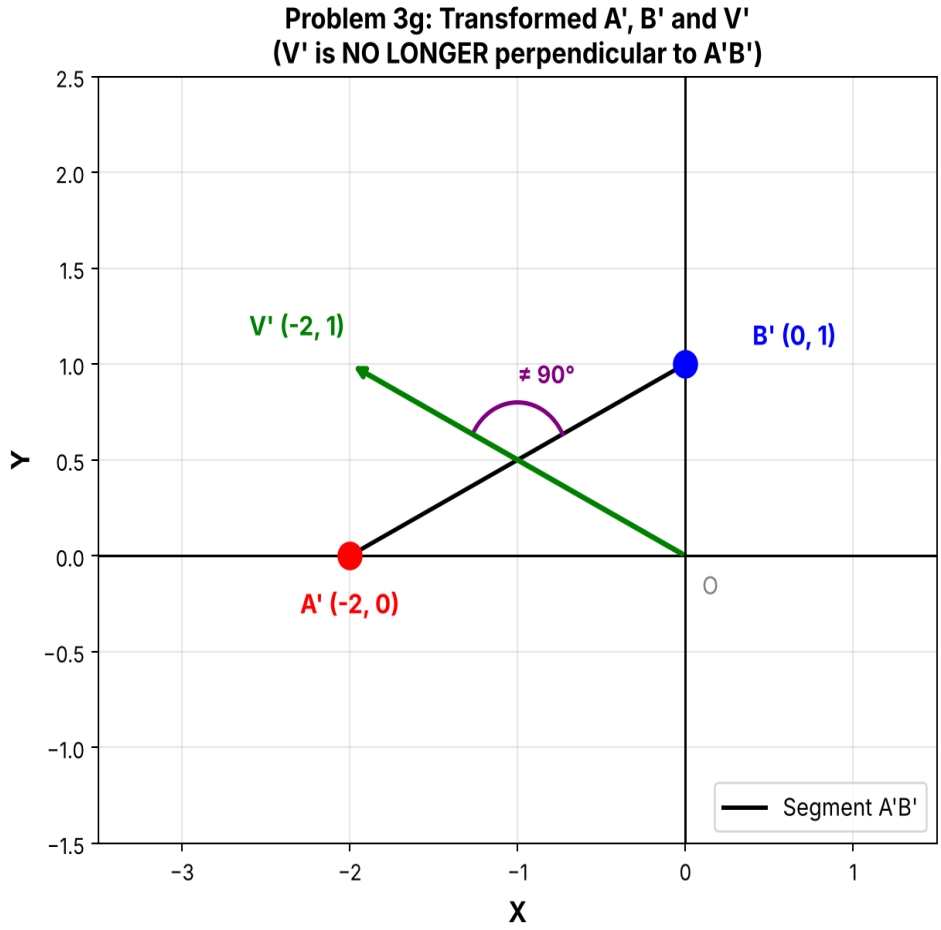
V' = M × V:

```
x' = 0(0.5) + (-4)(0.5) + 0(0) + 0(0) = -2.0
y' = 2(0.5) + 0(0.5) + 0(0) + 0(0) = 1.0
```

V' = [-2.0, 1.0, 0.0, 0]^T

(w' = 0 confirms V' is still a vector.)

3g) Sketch transformed elements A', B' and V'



A' is at (-2, 0), B' is at (0, 1). The segment A'B' and vector V' = (-2, 1) are shown. The angle marker shows these are no longer perpendicular.

3h) Is V' normal to line segment A'B'?

Step 1: Direction vector of A'B':

```
A'B' = B' - A' = [0-(-2), 1-0] = [2, 1]
```

Step 2: Dot product V' · A'B':

```
V' . A'B' = (-2)(2) + (1)(1) = -4 + 1 = -3
```

`Since V' . A'B' = -3 (NOT zero), V' is NOT normal to A'B'.`

Explanation: The original vector V was perpendicular to AB, but after applying M (which contains a non-uniform scaling S(2,4,1)), V' is no longer perpendicular to A'B'. Non-uniform scaling distorts angles because it stretches different axes by different amounts (2x in x, 4x in y). While pure rotations and uniform scaling preserve perpendicularity, non-uniform scaling does not.

This is why in computer graphics, surface normals must be transformed using the **inverse-transpose** of the model matrix rather than the model matrix itself, to maintain correct perpendicularity for lighting calculations.